

Report
Of
JSON Database Simulator

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.
In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 2410031637
Name of Student: GIRIK GOSWAMI
Batch: 2024-2028

CANDIDATE'S DECLARATION

I, GIRIK GOSWAMI do hereby declare that the internship report titled "JSON Database Simulator" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name : GIRIK GOSWAMI

Roll No : 2410031637

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

Student Name : GIRIK GOSWAMI

Roll No : 2410031637

Chapter No.	Chapter Name	Page Nos.
	Cover Page	----
1.	Candidate's Declaration	2
2.	Acknowledgment	3
3.	Internship Completion Certificate	5
4.	Project Description 4.1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of completion and other communication mail proof must be attached	6 6 6 7 7 7 8 8 9 9 10-12
5.	Bibliography/References	13

INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Girik Goswami

IILM University, Greater Noida

has successfully completed the 8-weeks

Python Full Stack Development With Project Virtual Internship

During June - August 2026

Supported by



EduSkills
ACADEMY

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4ff0aab66ab5a9ebf7c3

Student ID : STU69ecc320b22ec1777124128



4. PROJECT DESCRIPTION

4.1 Introduction

Internships play an important role in providing students with practical exposure and helping them understand the application of academic knowledge in real-world environments. The AICTE-EduSkills Virtual Internship provided hands-on exposure to Python Full Stack Development With Project and industry-oriented software development practices. The internship was conducted in an online mode through the EduSkills LMS, allowing the student to gain practical understanding of programming, data handling, and project development. During the internship, a project titled "JSON Database Simulator" was developed to apply the concepts and skills gained during the internship.

4.2 ORGANISATION PROFILE

- **Organization Name:** EduSkills Academy
- **Domain:** EdTech and Skill Development
- **Service:** Virtual Internships
- **Vision:** To empower students with industry-relevant skills.

EduSkills Academy provides virtual internship opportunities that help students gain practical exposure to industry-oriented concepts and applications. The internship was conducted online through the EduSkills LMS, with the student working in the role of an intern.

4.3 PROBLEM STATEMENT

The JSON Database Simulator is a project focused on building a functional, in-memory NoSQL database. This system is designed to emulate the behavior of a NoSQL database by processing various commands, including data creation, retrieval, modification, and deletion. It utilizes a text-based interface for user interaction and consistently returns results in the standardized JSON format. The simulator also incorporates advanced features for data filtering and generating snapshots, offering a comprehensive yet simplified environment for learning database management principles.

4.4 PROJECT OBJECTIVES

The objectives of the project include designing and implementing a solution for the given problem statement, utilizing modern tools and algorithms, and optimizing performance and code quality. The system provides a basic rule-based approach for database emulation that can be further extended by incorporating actual persistence or API integrations.

4.5 SCOPE OF PROJECT

The scope of the project includes establishing an in-memory data structure using Python dictionaries to represent the database. The system processes incoming string commands to perform core CRUD operations (Create, Read, Update, Delete). It utilizes Pythonic techniques, such as list comprehensions and lambda functions, to perform advanced functionalities like filtering and creating data snapshots, outputting all results strictly as formatted JSON.

4.6 TECHNOLOGIES AND TOOLS USED

The project uses the following technologies and concepts:

- **Programming Language:** Python
- **Library:** JSON for data handling (parsing and serialization)
- **Data Structures:** Python Dictionaries and Lists
- **Core Concepts:** In-memory databases, NoSQL principles, CRUD operations, Data filtering, Data snapshots.

These technologies and concepts are used for parsing activity data, processing user interactions, aggregating information, identifying preferences, and generating structured output.

4.7 SYSTEM ARCHITECTURE

The system follows a sequential data-processing and execution workflow:

User Command Input -> Command Parsing -> Identify Target Database -> Parse JSON Data Payload -> Execute Operation (CRUD) -> Format Output -> JSON Serialization.

In the first stage, raw user commands are passed into the system via standard input. The system identifies the action (CREATE, INSERT, UPDATE, DELETE, QUERY, SNAPSHOT) and targets the specific list of records. The internal database state is modified in-memory, and the operation's success or failure is appended to a results list, which is finally dumped as a JSON string.

4.8 METHODOLOGY

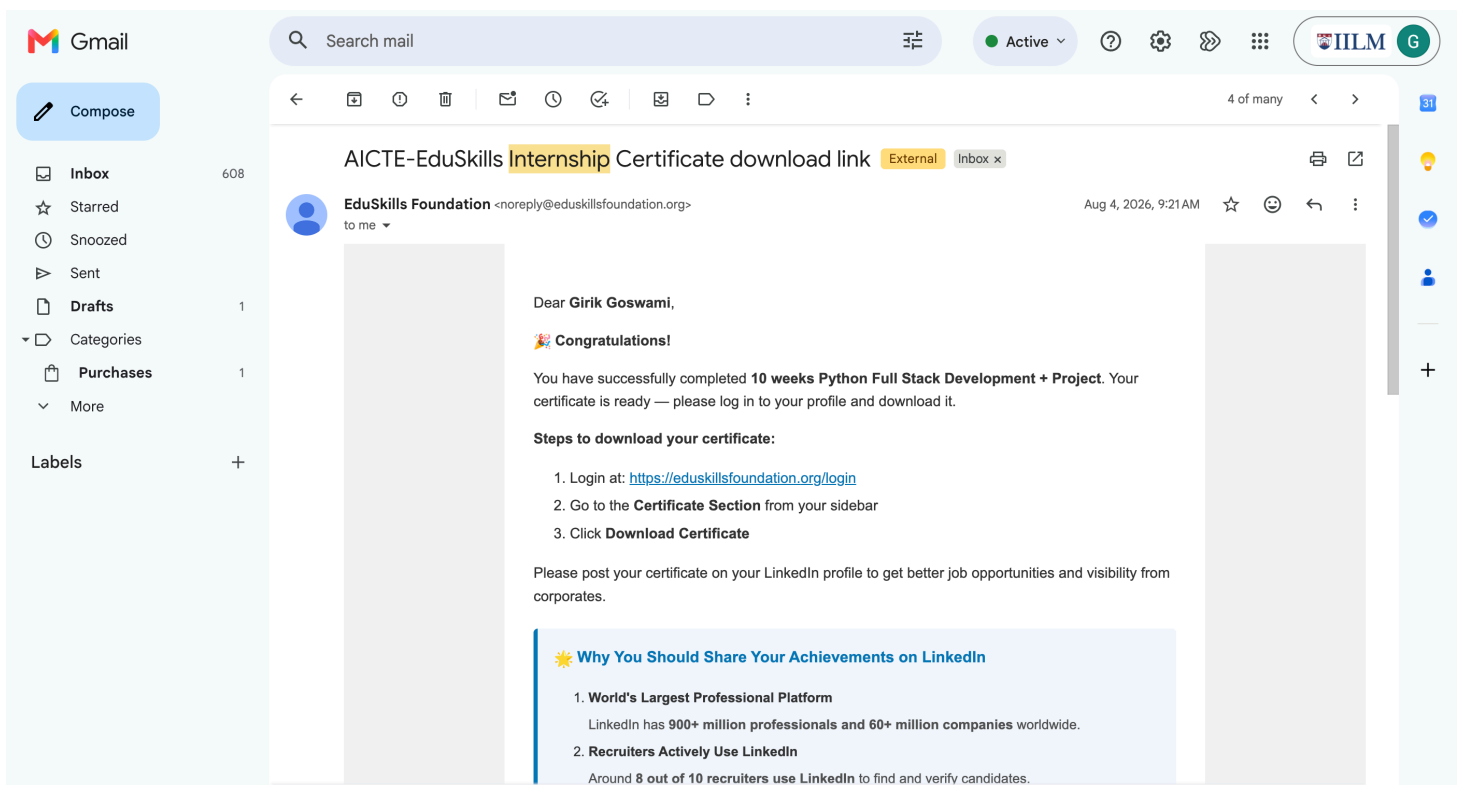
The project was implemented through a series of interconnected steps:

1. **Database Initialization:** Establishing the fundamental data structure using Python dictionaries to represent databases and lists to organize records.
2. **Inserting Records:** Parsing incoming JSON string data and appending the structured objects to the database's internal list.
3. **Updating Records:** Identifying specific records based on unique identifiers and altering their associated values while maintaining data integrity.
4. **Deleting Records:** Filtering out elements from the active data set that meet specified deletion criteria.
5. **Filtering and Snapshot:** Utilizing list comprehensions to retrieve subsets of data based on specific conditions and preserving the database state.

4.9 EXPECTED OUTCOME

The project successfully demonstrates the fundamental principles of database management within an in-memory, NoSQL context. By implementing core CRUD operations and advanced querying features, the system highlights practical applications of data persistence and manipulation. It effectively develops proficiency in handling JSON data, Pythonic data structures, and algorithmic efficiency while producing structured JSON output that can be used consistently for database results.

4.10 CERTIFICATE OF COMPLETION AND OTHER PROOFS



Ref No: C17Y2026M081483891

Internship Offer Letter

Student Name : Girik Goswami
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : Python Full Stack Development + Project
Internship Duration : June 2026 - August 2026
AICTE Student ID : STU69ecc320b22ec1777124128

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Introduction to Full Stack & Python Fundamentals	Introduction to full-stack development concepts and environment setup. Learn Python fundamentals including data types, functions, OOP concepts, and backend basics.
Week 2	FastAPI & Data Validation (Pydantic)	Build backend APIs using FastAPI framework. Learn request handling, routing, dependency injection, and data validation using Pydantic.
Week 3	React Fundamentals & Hooks	Learn React basics including components, JSX, props, state management, useState, useEffect, and building dynamic user interfaces.
Week 4	Reusable Components & Frontend-Backend Integration	Create reusable React components and integrate frontend with FastAPI backend using API calls and dynamic data rendering.
Week 5	Database Integration with SQLAlchemy	Integrate databases using SQLAlchemy ORM, perform CRUD operations, manage models, and connect FastAPI with relational databases.
Week 6	Advanced React & State Management	Learn advanced React patterns, routing with React Router, and state management using Redux or Zustand for scalable applications.
Week 7	Testing & Deployment	Implement backend and frontend testing strategies. Learn deployment techniques using Docker and cloud platforms.
Week 8	API design & Front end optimisation Capstone Project – Full Stack Application Development & Capstone Project	Begin building a complete full-stack application (E-commerce platform) integrating FastAPI backend, database (SQLAlchemy), authentication system, and React frontend.

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.



Bibek Ranjan
Director, EduSkills

Scan to Verify



Supported by



Under the Patronage of



Bibliography and References

1. Python Software Foundation, "Python Documentation," Python Documentation. Available at: <https://docs.python.org/3/>
2. JSON.org, "Introducing JSON," JSON Specification. Available at: <https://www.json.org/json-en.html>
3. Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser, "Data Structures and Algorithms in Python," Wiley.
4. Real Python, "Tutorials and articles on Python programming and data handling."
5. EduSkills Academy, AICTE-EduSkills Virtual Internship Program, Internship Project Report and Completion Certificate, 2026.